# Linear Algebra with Applications in Automatic Control

School of Automation,
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Fall Semester

# 3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)

Definition 2.7 The matrix $N_i$ given below is called the column companion matrix of the polynomial $\psi_i(\lambda) = \lambda^{n_i} + \alpha_{n_i-1,i}\lambda^{n_i-1} + \cdots + \alpha_{1,i}\lambda + \alpha_{0,i} \in \mathcal{F}[\lambda]$, where

$$N_i = \begin{bmatrix} 0 & 0 & \dots & 0 & -\alpha_{0,i} \\ 1 & 0 & \ddots & \vdots & -\alpha_{1,i} \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & 1 & -\alpha_{n_i-1,i} \end{bmatrix} \in \mathcal{F}^{\nu\times\nu}$$

Theorem 3.9 $A \in \mathcal{F}^{n\times n}$ has the nonconstant invariant polynomials $\psi_1(\lambda)$, $\psi_2(\lambda)$, ..., $\psi_l(\lambda)$ with $\deg \psi_i(\lambda) = n_i$, $\psi_i(\lambda)|\psi_{i+1}(\lambda)$, $i = 1, 2, \dots, l-1$, $N_i \in \mathcal{F}^{n_i\times n_i}$ is the column companion matrix of $\psi_i(\lambda)$. Then

$$A \sim N = \text{blockdiag}\{N_1, N_2, \dots, N_l\}. \quad (1)$$

- N is the natural normal form

# 3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Let $n_i = \deg \psi_i(\lambda),\ i = 1, 2, \ldots, l,$

$$\lambda E_n - A \cong \text{diag}\left\{ E_{n_1-1}, E_{n_2-1}, \ldots, E_{n_l-1}, \psi_1(\lambda), \psi_2(\lambda), \ldots, \psi_l(\lambda) \right\}.$$

$$\text{diag}\left\{ E_{n_1-1}, \psi_1(\lambda), E_{n_2-1}, \psi_2(\lambda), \ldots, E_{n_l-1}, \psi_l(\lambda) \right\}$$

Denote $S_i(\lambda) = \text{diag}\{ E_{n_i-1}, \psi_i(\lambda) \}$, then

$$\lambda E_n - A \cong \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \}.$$

It is clear that $S_i(\lambda) \in \mathcal{F}^{n_i \times n_i}[\lambda]$ is a diagonal $\lambda$-matrix and its only nonconstant diagonal element $\psi_i(\lambda)$ is of degree $n_i$ and at the bottom. Hence, $S_i(\lambda)$ is in Smith canonical form with the only nonconstant invariant polynomial $\psi_i(\lambda)$.

# 3.2 The Natural normal form

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- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Recalling example 2.26 and the symmetry of the equivalence of $\lambda$-matrices, we obtain

$$S_i(\lambda) \cong \lambda E_{n_i} - N_i$$

where

$$N_i = \begin{bmatrix} 0 & 0 & \dots & 0 & -\alpha_{0,i} \\ 1 & 0 & \ddots & \vdots & -\alpha_{1,i} \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & 1 & -\alpha_{n_i-1,i} \end{bmatrix} \in \mathcal{F}^{\nu\times\nu} \quad (1)$$

with

$$\psi_i(\lambda) = \lambda^{n_i} + \alpha_{n_i-1,i}\lambda^{n_i-1} + \cdots + \alpha_{1,i}\lambda + \alpha_{0,i}\,,$$

# 3.2 The Natural normal form

Z. Song

- Natural normal form via similar transformation (reduction)
  - Derivation is on Pg 96, the key points:

Hence

$$\begin{aligned} & \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \} \\ \cong\ & \text{blockdiag}\{ \lambda E_{n_1} - N_1, \lambda E_{n_2} - N_2, \ldots, \lambda E_{n_l} - N_l \} \end{aligned}$$

Denote

$$N = \text{diag}\{ N_1, N_2, \ldots, N_l \}$$

then from the transitivity of the equivalence of $\lambda$-matrices follows

$$\lambda E_n - A \cong \text{blockdiag}\{ S_1(\lambda), S_2(\lambda), \ldots, S_l(\lambda) \} \cong \lambda E_n - N$$

From theorem 3.8 we have $A \sim N$.
Since $N$ is block diagonal and the block diagonals have the simple structure the form of $N_i$, we have thus found a simple form for $A$ in its similarity class.

# 3.3 The Jordan Canonical forms

Z. Song

- Based on the noconstant invariant polynomials, we give the natural form,

- Nonconstant invariant polynomial can be factorized into powers of comprime factors (primes),

- based on this fact we introduce another simple form for A, Jordan canonical form

- Recall factorization of polynomials in the field of C, and R,

- The primes are of with degree one in C,

- The primes are of with degree one or two in R, are symmetry with respect to the real axis

- Depending on how we factorize the invairant polynomials of $(\lambda E_n - A)$ ,we give two kinds of Jordan forms for A, complex case, and real case

# 3.3 The Jordan Canonical forms

Z. Song

## •3.3.1 Jordan form in the complex matrices case

•First introduce an important lemma which lays the ground for Jordan form

Lemma 3.1 Let $\mathcal{F}$ be a field,

$C(\lambda) = \text{diag}\{\varphi_1(\lambda), \varphi_2(\lambda), \ldots, \varphi_s(\lambda)\} \in \mathcal{F}^{s\times s}[\lambda]$

with g.c.d$\{\varphi_j(\lambda), \varphi_k(\lambda)\} = 1\ \forall j \neq k$. Then

$$C(\lambda) \cong \text{diag}\left\{E_{s-1}, \prod_{j=1}^{s} \varphi_j(\lambda)\right\}$$

Since $\mathcal{C}$ is algebraically closed, any nonconstant invariant polynomial of $\det(\lambda E_n - A)$ or $(A)$ for short

$$\psi_i(\lambda) = \lambda^{n_i} + \alpha_{n_i-1,i}\lambda^{n_i-1} + \cdots + \alpha_{1,i}\lambda + \alpha_{0,i} \in \mathcal{C}[\lambda]$$

can be factorized as

$$\psi_i(\lambda) = (\lambda - \lambda_1)^{n_{i1}}(\lambda - \lambda_2)^{n_{i2}} \cdots (\lambda - \lambda_s)^{n_{is}}$$

where $\lambda_j \in \mathcal{C}$, $\lambda_j \neq \lambda_k\ \forall j \neq k$, $n_{ij} \geq 0$, $j = 1, 2, \ldots, s$, satisfying $n_i = \sum_{j=1}^{s} n_{ij}$, hence at least one of $n_{ij}$ is positive.
It is clear that $\lambda_j$, $j = 1, 2, \ldots, s$, are the all distinct eigenvalues of $A$. In Lemma 3.1, let $\varphi_j(s) = (\lambda - \lambda_j)^{n_{ij}}$, we have the Jordan form shown in the next slide.

# 3.3 The Jordan Canonical forms

Z. Song

## •3.3.1 Jordan form in the complex matrices case

Theorem 3.10 Let $N_i \in \mathcal{C}^{n_i \times n_i}$ be the column companion matrix of the $i$th nonconstant invariant polynomial $\psi_i(\lambda)$, and $\psi_i(\lambda)$ have the factorization in the last slide with $n_i = \sum_{j=1}^{s} n_{ij}$, then

1.

$$\begin{aligned} (\lambda E_{n_i} - N_i) &\cong \text{diag}\{ T_{i1}(\lambda), T_{i2}(\lambda), \ldots, T_{is}(\lambda) \} \\ \text{where} \quad T_{ij}(\lambda) &= \text{diag}\left\{ E_{n_{ij}-1}, (\lambda - \lambda_j)^{n_{ij}} \right\} \in \mathcal{C}^{n_{ij} \times n_{ij}}[\lambda] \end{aligned}$$

2.

$$N_i \sim J_i = \text{blockdiag}\{ J_{i1}, J_{i2}, \ldots, J_{is} \},$$

where

$$J_{ij} = \lambda_j E_{n_{ij}} + H_{n_{ij}} = \begin{bmatrix} \lambda_j & 1 & 0 & \ldots & 0 \\ 0 & \lambda_j & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ldots & 0 & \lambda_j & 1 \\ 0 & \ldots & 0 & 0 & \lambda_j \end{bmatrix} \in \mathcal{C}^{n_{ij} \times n_{ij}}$$

# 3.3 The Jordan Canonical forms



## •3.3.1 Jordan form in the complex matrices case

Theorem 3.11 Let $A \in \mathcal{C}^{n\times n}$ have the non-constant invariant polynomials $\psi_1(\lambda)$, $\psi_2(\lambda)$, ..., $\psi_l(\lambda)$ with the coprime factorizations

$$\begin{aligned}
\psi_1(\lambda) &= (\lambda-\lambda_1)^{n_{11}}(\lambda-\lambda_2)^{n_{12}}\cdots(\lambda-\lambda_s)^{n_{1s}} \\
\psi_2(\lambda) &= (\lambda-\lambda_1)^{n_{21}}(\lambda-\lambda_2)^{n_{22}}\cdots(\lambda-\lambda_s)^{n_{2s}} \\
&\vdots \\
\psi_l(\lambda) &= (\lambda-\lambda_1)^{n_{l1}}(\lambda-\lambda_2)^{n_{l2}}\cdots(\lambda-\lambda_s)^{n_{ls}}
\end{aligned} \tag{1}$$

then

$$A \sim \text{blockdiag}\{J_1, J_2, \ldots, J_l\} \tag{2}$$

$$J_i = \text{blockdiag}\{J_{i1}, J_{i2}, \ldots, J_{is}\} \tag{3}$$

$$J_{ij} = \begin{bmatrix} \lambda_j & 1 & 0 & \ldots & 0 \\ 0 & \lambda_j & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \ldots & 0 & \lambda_j & 1 \\ 0 & \ldots & \ldots & 0 & \lambda_j \end{bmatrix} \in \mathcal{C}^{n_{ij}\times n_{ij}} \tag{4}$$

# 3.3 The Jordan Canonical forms



- 3.3.1 Jordan form in the complex matrices case

Definition 3.9 The $n_{ij} \times n_{ij}$ matrix $J_{ij}$ in (4) of the last slide is called a Jordan block associated with the eigenvalue $\lambda_j$. The block diagonal matrix

$$J = \text{blockdiag}\{J_1 \ , \ J_2 \ , \ \ldots \ , \ J_l\}$$

given in the equations (2) through (4) is called the Jordan canonical form of the matrix $A$.

- Let us have a look at one example, Ex. 3.5 on Pg 103.

# 3.3 The Jordan Canonical forms

Z. Song

## •3.3.2 Jordan form in the real matrices case

•Invariant polynomials are factorized into powers of primes with degree one and two, Theorem 2.17, real roots, complex roots appearing pairwise

Consider $n_{ik} \times n_{ik}$ complex Jordan blocks

$$J_{ik}(\lambda_k) = \begin{bmatrix} \lambda_k & 1 & 0 & \dots & 0 \\ 0 & \lambda_k & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & \lambda_k & 1 \\ 0 & \dots & 0 & 0 & \lambda_k \end{bmatrix},$$

$$J_{ik}(\bar{\lambda}_k) = \begin{bmatrix} \bar{\lambda}_k & 1 & 0 & \dots & 0 \\ 0 & \bar{\lambda}_k & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & \bar{\lambda}_k & 1 \\ 0 & \dots & 0 & 0 & \bar{\lambda}_k \end{bmatrix},$$

$$J_{n_{ik}}(\lambda_k, \bar{\lambda}_k) = \begin{bmatrix} I(\sigma_k, \omega_k) & E_2 & 0 & \dots & 0 \\ 0 & I(\sigma_k, \omega_k) & E_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & I(\sigma_k, \omega_k) & E_2 \\ 0 & \dots & 0 & 0 & I(\sigma_k, \omega_k) \end{bmatrix} \in \mathcal{R}^{(2n_{ik})\times(2n_{ik})}.$$

where

$$I(\sigma_k, \omega_k) = \begin{bmatrix} \sigma_k & \omega_k \\ -\omega_k & \sigma_k \end{bmatrix}.$$

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.3 Find the transition matrix X, such inv(X)AX = J
- Next introduce generalized eigenvector, find X, such that AX=XJ

Definition 3.10 Let $A \in \mathcal{C}^{n\times n}$, $\lambda_j$ be an eigenvalue of $A$. $\boldsymbol{x} \in \mathcal{C}^n$ is referred to as a generalized eigenvector of order $k$ associated with $\lambda_j$ if it satisfies the following conditions

$$\begin{cases} (A-\lambda_j E_n)^{k-1}\boldsymbol{x} \neq \boldsymbol{0}\,, \\ (A-\lambda_j E_n)^{k}\boldsymbol{x} = \boldsymbol{0}\,. \end{cases} \tag{1}$$

Definition 3.11 Let $A \in \mathcal{C}^{n\times n}$, $\lambda_j$ be an eigenvalue of $A$. The sequence of vectors $[\boldsymbol{x}_1 \ \boldsymbol{x}_2 \ \ldots \ \boldsymbol{x}_l]$ satisfying

$$\begin{cases} (A-\lambda_j E_n)\boldsymbol{x}_{k+1} = \boldsymbol{x}_k & k = 1,2,\ldots,l-1 \\ \boldsymbol{x}_1 \in \mathcal{N}(A-\lambda_j E_n)\,, & \boldsymbol{x}_l \notin \mathcal{R}(A-\lambda_j E_n)\,. \end{cases}$$

is called a Jordan chain of length $l$ associated with $\lambda_j$. $\boldsymbol{x}_1$ is called the initial and $\boldsymbol{x}_l$ is called the terminate.

# 3.3 The Jordan Canonical forms



- 3.3.3 Find the transition matrix X, such inv(X)AX = J

- AX=XJ

Theorem 3.13 Let $A \in \mathcal{C}^{n\times n}$, $\lambda_1, \lambda_2, \ldots, \lambda_s$ be the distinct eigenvalues of $A$. $\lambda E_n - A$ has $q_j$ elementary divisors in the form of powers of $(\lambda - \lambda_j)$ with the power indices

$$0 < n_{l-q_j+1,j} \le n_{l-q_j+2,j} \le \cdots \le n_{lj}\,.$$

Then $A$ has $q_j$ Jordan chains associated with $\lambda_j$ of lengths $n_{l-q_j+1,j}, n_{l-q_j+2,j}, \ldots, n_{lj}$.

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.3 Find the transition matrix X, such inv(X)AX = J

- AX=XJ

Theorem 3.13 Let $A \in \mathcal{C}^{n\times n}$, $\lambda_1, \lambda_2, \ldots, \lambda_s$ be the distinct eigenvalues of $A$. $\lambda E_n - A$ has $q_j$ elementary divisors in the form of powers of $(\lambda - \lambda_j)$ with the power indices

$$0 < n_{l-q_j+1,j} \le n_{l-q_j+2,j} \le \cdots \le n_{lj}\,.$$

Then $A$ has $q_j$ Jordan chains associated with $\lambda_j$ of lengths $n_{l-q_j+1,j}, n_{l-q_j+2,j}, \ldots, n_{lj}$.

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.3 Find the transition matrix X, such inv(X)AX = J

- AX=XJ, X is invertible

Theorem 3.14 Let $\boldsymbol{x}_{ij,1}, \boldsymbol{x}_{ij,2}, \ldots, \boldsymbol{x}_{ij,n_{ij}}$ be a Jordan chain associated with the eigenvalue $\lambda_j$. Then the vectors $\boldsymbol{x}_{ij,1}, \boldsymbol{x}_{ij,2}, \ldots \boldsymbol{x}_{ij,n_{ij}}$ are linearly independent.

Theorem 3.15 Let $X_{ij} = [\boldsymbol{x}_{ij,1} \ \boldsymbol{x}_{ij,2} \ \ldots \ \boldsymbol{x}_{ij,n_{ij}}]$ and $X_{kj} = [\boldsymbol{x}_{kj,1} \ \boldsymbol{x}_{kj,2} \ \ldots \ \boldsymbol{x}_{kj,n_{kj}}]$ be two Jordan chains associated with $\lambda_j$, $\boldsymbol{x}_{ij,1}$ and $\boldsymbol{x}_{kj,1}$ are the two initials. The columns in

$$[X_{ij} \ \ X_{kj}] = [\boldsymbol{x}_{ij,1} \ \boldsymbol{x}_{ij,2} \ \ldots \ \boldsymbol{x}_{ij,n_{ij}} \ \boldsymbol{x}_{kj,1} \ \boldsymbol{x}_{kj,2} \ \ldots \ \boldsymbol{x}_{kj,n_{kj}}]$$

are linearly independent, if and only if $\boldsymbol{x}_{ij,1}$ and $\boldsymbol{x}_{kj,1}$ are linearly independent.

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.3 Find the transition matrix X, such inv(X)AX = J

- AX=XJ, X is invertible-continued

Theorem 3.16 Let $\lambda_j \neq \lambda_k$ be two distinct eigenvalues of $A \in \mathcal{C}^{n\times n}$ with the Jordan chains $X_i = [\boldsymbol{x}_{i1} \quad \boldsymbol{x}_{i2} \quad \dots \quad \boldsymbol{x}_{in_i}]$, where $i = j, k$. Then the column of the matrix $[X_j \quad X_k]$ are linearly independent.

Theorem 3.17 Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be eigenvalues of $A \in \mathcal{C}^{n\times n}$ (not necessarily distinct), the columns of $X_{ij} = [\boldsymbol{x}_{ij,1} \quad \boldsymbol{x}_{ij,2} \quad \dots \quad \boldsymbol{x}_{ij,n_{ij}}]$ be a Jordan chain associated with $\lambda_j$, $i = 1, 2, \dots, k$, $j = 1, 2, \dots, m$. Then $[X_{11} \quad X_{21} \quad \dots \quad X_{km}]$ has linearly independent columns if and only if the initials $\boldsymbol{x}_{11,1}, \boldsymbol{x}_{21,1}, \dots, \boldsymbol{x}_{km,1}$ are linearly independent.

# 3.3 The Jordan Canonical forms

Z. Song

## •3.3.4 Decomposing C^{n} into direct sum of Jordan subspaces of A

Theorem 3. 18 Let $X \in \mathcal{C}_n^{n\times n}$ be the transition matrix such that $A \in \mathcal{C}^{n\times n}$ is reduced to Jordan canonical form, i.e.

$$\begin{aligned} X^{-1}AX &= J = \text{blockdiag}\{J_1, J_2, \ldots, J_l\} \\ J_i &= \text{blockdiag}\{J_{i1}, J_{i2}, \ldots, J_{is}\} \\ J_{ij} &= \lambda_j E_{n_{ij}} + H_{n_{ij}}. \end{aligned}$$

Decompose $X$ according to the dimensions of $J_i$ into $l$ block columns $X = [X_1 \quad X_2 \quad \ldots \quad X_l]$ with $l$ being the number of nonconstant invariant polynomials of $A$, and decompose further $X_i$ according to the dimensions of $J_{ij}$ into at most $s$ block columns $X_i = [X_{i1} \quad X_{i2} \quad \ldots \quad X_{is}]$ with $s$ being the number of distinct eigenvalues of $A$, then

$$\mathcal{T}(\lambda_j) = \text{span}\{X_{1j}\} \dotplus \text{span}\{X_{2j}\} \dotplus \cdots \dotplus \text{span}\{X_{lj}\}, \tag{1}$$

where $\text{span}\{X_{ij}\}$ is the subspace spanned by the column vectors of $X_{ij}$.

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.4.1 Decomposing C^{n} into direct sum of Jordan subspaces of A

  - Define Jordan subspace

Let $\lambda_j$ be an eigenvalue of $A \in \mathcal{C}^{n\times n}$,

$$\mathcal{T}(\lambda_j) \triangleq \left\{ \boldsymbol{x} \ : \ \text{there exists } k \in \mathcal{Z}_+ \text{such that } (A-\lambda_j E_n)^k \boldsymbol{x} = \boldsymbol{0} \right\}. \tag{1}$$

Definition 3.12 $\mathcal{T}(\lambda_j)$ is called the Jordan subspace associated with the eigenvalue $\lambda_j$ of $A$.

# 3.3 The Jordan Canonical forms

Z. Song

- 3.3.4.1 Decomposing C^{n} into direct sum of Jordan subspaces of A

Theorem 3.18 reveals that the Jordan subspace $\mathcal{T}(\lambda_j)$ associated with $\lambda_j$ is the direct sum of the $q_j$ subspaces spanned by the column vectors in the Jordan chains

$$\mathcal{T}(\lambda_j) = \text{span}\{X_{1j}\} \dot{+} \text{span}\{X_{2j}\} \dot{+} \cdots \dot{+} \text{span}\{X_{lj}\}. \tag{1}$$

Thus given $A \in \mathcal{C}^{n\times n}$, $\mathcal{C}^n$ can be decomposed into the direct sum of $s$ Jordan subspaces associated with the distinct eigenvalues of $A$

$$\mathcal{C}^n = \mathcal{T}(\lambda_1) \dot{+} \mathcal{T}(\lambda_2) \dot{+} \cdots \dot{+} \mathcal{T}(\lambda_s). \tag{2}$$

# 3.3 The Jordan Canonical forms



- 3.3.4.2 Decomposing V into direct sum of Jordan subspaces associated with the linear transformation *A*

Define

$$\left[\hat{\boldsymbol{\alpha}}_{ij,1} \quad \hat{\boldsymbol{\alpha}}_{ij,2} \quad \dots \quad \hat{\boldsymbol{\alpha}}_{ij,n_{ij}}\right] = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\, X_{ij}\,,$$

then, corresponding to (3.54), the Jordan subspace associated with the eigenvalue $\lambda_j$ of $\mathscr{A}$ is

$$\begin{aligned}\mathcal{V}(\lambda_j) \;=\; & \operatorname{span}\left\{\hat{\boldsymbol{\alpha}}_{1j,1}\,,\ \hat{\boldsymbol{\alpha}}_{1j,2}\,,\ \dots\,,\ \hat{\boldsymbol{\alpha}}_{1j,n_{1j}}\right\} \dot{+} \\ & \operatorname{span}\left\{\hat{\boldsymbol{\alpha}}_{2j,1}\,,\ \hat{\boldsymbol{\alpha}}_{2j,2}\,,\ \dots\,,\ \hat{\boldsymbol{\alpha}}_{2j,n_{2j}}\right\} \dot{+} \cdots \dot{+} \\ & \operatorname{span}\left\{\hat{\boldsymbol{\alpha}}_{lj,1}\,,\ \hat{\boldsymbol{\alpha}}_{lj,2}\,,\ \dots\,,\ \hat{\boldsymbol{\alpha}}_{lj,n_{lj}}\right\}. \qquad (1)\end{aligned}$$

In accordance with (3.55), $\mathcal{V}$ can be decomposed into the direct sum of the $s$ Jordan subspaces associated with the distinct eigenvalues of $\mathscr{A}$

$$\mathcal{V} = \mathcal{V}(\lambda_1) \dot{+} \mathcal{V}(\lambda_2) \dot{+} \cdots \dot{+} \mathcal{V}(\lambda_s)\,. \qquad (2)$$

Obviously, for $n_{ij} > 0$, $\hat{\boldsymbol{\alpha}}_{ij,1}$, $i = l - q_j + 1,\ l - q_j + 2,\ \dots,\ l$ are the eigenvectors of $\mathscr{A}$ associated with $\lambda_j$, $\hat{\boldsymbol{\alpha}}_{ij,k_i}$, $i = l - q_j + 1, i = l - q_j + 2, \dots, l$, $k_i = 1, 2, \dots, n_{ij}$ are the successive vectors in the $i$th Jordan chain associated with the eigenvalue $\lambda_j$ of $\mathscr{A}$.

# 3.3 The Jordan Canonical forms



- Decomposing V, please refer to Pg 115, the bottom.

If $A$ is the representation of a linear transformation $\mathscr{A}$ on the linear space $\mathcal{V}$ over $\mathcal{C}$ with respect to the basis $\alpha_1, \alpha_2, \ldots, \alpha_n$, then $\mathcal{V}$ can be accordingly decomposed.

- This is the end of Jordan decomposition